

# SupplyBridge Backend Implementation Plan

## **Phase 1: Backend Setup + MySQL + Prisma + Authentication + RBAC**

Create backend folder, Express server, environment configuration, MySQL connection, Prisma ORM setup, migrations, seed data, JWT authentication, password hashing, role-based access control (RBAC), roles, permissions, middleware, user management, initial REST APIs, testing of authentication and permissions.

## **Phase 2: Products, Suppliers, Stores, Catalog, Data Mapping**

Implement Product Information Management (PIM), suppliers, supplier onboarding, manufacturers, brands, categories, variants, media library, multi-store management, store settings, supplier connections (API/FTP/SFTP/SOAP), mapping modules (product/category/brand/variant/attribute), CRUD APIs and database integration.

## **Phase 3: Validation Center, Sync Engine, Queue Management, Dashboard, Logs, Testing**

Implement validation engine (duplicate SKU, missing images, pricing, inventory etc.), sync engine, background jobs, queue management, retry/pause/resume/cancel, dashboard statistics, activity logs, sync logs, API logs, import logs, system settings, end-to-end testing, API testing, database testing, production readiness.